

2026 GLOBAL QUANTUM + AI CHALLENGE

Assessment Criteria

Phase 1 (Ideation) and Phase 2 (PoC Sprint) - For Reviewers and Participants

1. Introduction

This document sets out the assessment criteria used to evaluate submissions to the 2026 Global Quantum + AI Challenge. The criteria are published for transparency and to help participants understand what reviewers are looking for. The same criteria framework applies across all enterprise problem statements, though reviewers may weight domain-specific considerations within each criterion.

The Challenge spans multiple sectors — healthcare (Cleveland Clinic), energy (E.ON), financial services (HSBC), aerospace (Airbus), and automotive/AI (Volkswagen Group) — and evaluates quantum, quantum-inspired, and quantum-AI hybrid approaches. The criteria are designed to be domain-agnostic at the category level while allowing reviewers to apply sector-relevant judgment.

2. Phase 1 Assessment Criteria (Ideation)

Each Phase 1 submission is scored on six criteria. Each criterion is scored on a 1–5 scale (1 = weak, 5 = exceptional). The total score is a weighted sum.

Criterion	What reviewers assess	Score	Weight
Problem Relevance & Impact	Does the proposal clearly address the enterprise problem statement? Is the potential impact meaningful and well-articulated? Would a successful outcome advance the use case in a tangible way?	1–5	25%
Technical Approach & Innovation	Is the proposed method technically sound and clearly explained? Does it offer a novel or differentiated approach? Is the quantum / quantum-AI framing credible (not superficial)?	1–5	25%
Feasibility	Can the proposed approach realistically be demonstrated within a 3–4 month PoC sprint? Are resource requirements (hardware, data, compute) realistic? Are assumptions and constraints clearly stated?	1–5	20%

Validation Plan	Is there a clear plan to test and measure results? Are success metrics defined? Is the benchmarking approach credible?	1–5	15%
Hybrid / Cross-Domain Approach	Does the proposal thoughtfully integrate quantum methods with classical, HPC, or AI workflows? Is there a clear rationale for the hybrid design?	1–5	5%
Team Capability	Does the team have the relevant technical expertise and domain knowledge? Is the team composition appropriate for the proposed work?	1–5	10%

Weighted total score: maximum 5.00.

3. Scoring Guide

Score	Rating	Description
5	Exceptional	Outstanding in all respects. Clear, compelling, and highly credible. Would be a flagship demonstration.
4	Strong	Convincing proposal with minor gaps. High confidence in feasibility and impact.
3	Competent	Solid but with notable weaknesses. Approach is reasonable but may lack differentiation or clarity.
2	Developing	Significant gaps in approach, feasibility, or relevance. Needs substantial rework.
1	Weak	Does not meet minimum expectations. Fundamental issues with approach, understanding, or relevance.

4. Phase 2 Assessment Criteria (PoC Sprint)

Phase 2 finalists are evaluated on the quality of their PoC demonstrators. The criteria shift from concept to execution. The below table is subject to change prior to the launch of Phase 2 but is provided for general guidance.

Criterion	What reviewers assess	Score	Weight
PoC Quality & Results	Does the PoC produce meaningful results? Are outputs well-documented and reproducible? Is there evidence of quantum advantage, parity, or a credible path to advantage?	1–5	30%
Enterprise Relevance	How well does the PoC address the original enterprise problem statement? Could the approach realistically be adopted or further developed by the enterprise?	1–5	25%
Technical Rigour	Is the methodology sound? Are benchmarks appropriate? Is the workflow clearly documented and reproducible?	1–5	20%
Scalability & Path Forward	Is there a credible roadmap from PoC to production? Are scalability constraints honestly assessed?	1–5	15%
Presentation & Communication	Is the work clearly presented? Can a non-specialist understand the significance? Quality of final pitch and documentation.	1–5	10%

5. Review Process and Integrity

5.1 Reviewer Assignment

Each Phase 1 submission is expected to be assessed by at least two reviewers. Reviewers are drawn from enterprise sponsors, technical experts (academic and industry), and assessors selected by Resonance.

5.2 Conflict of Interest

All reviewers are required to declare conflicts of interest to Resonance before reviewing any submission. A reviewer will be recused from any submission involving their own organization, a direct competitor, a financial interest, or a personal relationship with a team member. Resonance maintains a COI register and manages reviewer assignments accordingly.

5.3 Moderation

After individual scoring is complete, a moderation panel convenes to normalize scores, resolve significant disagreements between reviewers, and finalize the ranked list. The moderation panel does not re-score submissions but may adjust where scoring discrepancies indicate misalignment in interpretation.

5.4 Jury (Phase 2 Finals)

Phase 2 finalists should anticipate presenting to a jury comprising senior leaders from industry, academia, and the investment community. The jury composition will be announced publicly once confirmed. Jury members follow the same COI protocols as Phase 1 reviewers.